KiranAcademy

javabyKiran

DATA SCIENCE & Machine Learning using Python

Full Stack Development + Data Analytics

Teaching Languages

हिंदी | English

Thundercube Pvt. Ltd.

KiranAcademy

About Our Founder

Mr. Kiran Digrase

Founder and CEO of "The Kiran Academy"

14 years of experience in the **financial IT domain.**

Worked with **clients**

Why our Course?

- **155%** Highest Hike
- **₹16 Lakhs** Highest Salary
- **40% Hike** In Demand
- **738+** Hiring Partner
- **315+ Best** Industry Experts
- **100% Placement** Assurance

Our Success Journey

2014 Established **"Java By Kiran"** institute at **Karve Nagar, Pune.**

2018 We Become **"The Kiran Academy"** From **"Java By Kiran"**

2020 Opened our 2nd Branch at **Pimpri Chinchwad**

2021 Opened our 3rd Branch at **Warje, Pune.**

2022 Opened our 4th Branch at **Deccan, Pune.**

2023 Opened our 5th Branch at **Chatrapati Squar, Nagpur.**

2024 Opened our 5th Branch at **Hadapsar, Pune.**

About us

Vision

We bridge the gap by providing skilled professional training to students and helping them get placed in companies.

Our Mission

We empower individuals to upskill for IT software companies. We provide comprehensive **Industry focused** live training.

1. Contact us through our website or visit to the institute
2. Fill out the registration form & career counseling
3. Take admission & collect admission kit
4. Join WhatsApp broadcast message group
5. Join our learning portal
6. Attend online/classroom sessions
7. Take as many mock tests as you need
8. Prepare for the interview alongside a grooming session
9. Participate in job interviews & engage in interview rounds
10. Get placed & secure a job offer

Our Certifications

100% JOB ASSURANCE

25,000+ Students Placed

Best Placement Ratio

Common Features

Test, Assessments & Mocks

- Topic-wise Weekly Test
- Topic-wise Assessments
- Mock Interview Preparation
- One-On-One Discussion
- Interview Feedback On Emails

Best Practices & Coding Standards

- Naming Conventions
- Code Formatting & Style
- Writing Maintainable & Readable Code
- Code Reusability
- Code Auto-generation
- Real Time Scenarios
- Problem Statements & Solutions

Soft Skills

- 360° Development
- Resume Building
- Enhancing Confidence
- Logic Building Session
- English Communication

Debugging & Troubleshooting

- Use of Debugging Tools
- Identifying & Fixing Common Errors
- Handling Industrial Scenarios

What Makes Us Distinct?

- Exclusive Recorded Sessions
- Flexible Batch Switching Option
- Unlimited Mock Interviews
- 24/7 Live Support
- Personalized One-On-One Discussions
- 100% Transparency
- Comprehensive Technology Packages
- Industry Expert Trainers

INDEX

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Back-end Core Python

1. Introduction to Python Programming

- Capabilities of Python:
- Why Choose Python?
- Useful Insights:
- Python Syntax vs. Other Programming Languages
- Python Installation

2. Starting with Python Fundamentals

- Print Statements
- Comments
- Python Data Structures & Data Types
- String Operations in Python
- Simple Input & Output
- Simple Output Formatting
- Operators in Python

3. Python Program Flow Overview

- Indentation
- The If Statement and Its Related Statements
- Example with If and Its Related Statements
- The While Loop
- The For Loop
- The Range Statement
- Break & Continue
- Assert
- Looping Examples

4. Functions and Modules

- Creating Your Own Functions
- Function Parameters
- Variable Arguments
- Scope of a Function
- Function Documentation
- Lambda Functions & Map
- Exercise with Functions
- Creating a Module
- Standard Modules

5. Exception Handling in Python

- Errors
- Exception Handling with Try
- Handling Multiple Exceptions
- Writing Your Own Exception

6. File Handling

- File Handling Modes
- Reading Files
- Writing & Appending to Files
- Handling File Exceptions
- The 'with' Statement

7. Classes in Python

- New Style Classes
- Creating Classes
- Instance Methods
- Inheritance
- Polymorphism
- Exception Classes & Custom Exceptions

8. Generators and Iterators

- Iterators
- Generators
- Functions 'any' and 'all'
- 'with' Statement
- Data Compression

9. Data Structures

- Module
- List Comprehensions
- Nested List Comprehensions
- Dictionary Comprehensions
- Functions
  - Default Parameters
  - Variable Arguments
  - Specialized Sorts

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10. Collections

- Module
- namedtuple()
- deque
- ChainMap
- Counter
- OrderedDict
- defaultdict
- UserDict
- UserList
- UserString

11. Creating GUIs with Python (Tkinter)

- Introduction
- Components and Events
- An Example GUI
- The Root Component
- Adding a Button
- Entry Widgets
- Text Widgets
- Check Buttons

12. Python SQL Database Access

- Module Introduction
- Installation
- DB Connection
- Creating DB Table
- INSERT, READ, UPDATE, DELETE Operations
- COMMIT & ROLLBACK Operations
- Handling Errors

13. Date and Time

- Module
- sleep
- Program Execution Time
- Additional Methods on Date/Time

14. Exploring Additional Topics in Detail

- Filter
- Map
- Reduce
- Decorators
- Collections

15. Regular Expressions

1. Introduction to Regular Expressions
   - Definition and importance
   - Installing and using the re module
   - Basic syntax and pattern rules
2. Splitting Strings with Regex
   - Using split() for basic and advanced splitting
   - Handling multiple delimiters
3. Working with Special Characters, Dates, and Emails
   - Special characters (\d, \w, \s, .)
   - Extracting and validating dates (DD/MM/YYYY, YYYY-MM-DD)
   - Finding and validating emails
4. Using Quantifiers in Regex
   - *, +, ?, {n}, {n,}, {n,m}
   - Extracting numbers, validating input formats
5. Match and Find All in Python
   - match(), search(), and findall() usage
   - Extracting multiple occurrences of a pattern
6. Character Sequence and Substitute
   - Using character sets [ ]
   - re.sub() for replacing text
7. Using the Search Method in Regex
   - Finding first occurrence with search()
   - Extracting and locating matches

16. Essentials of Threads

- Class and Threads
- Multi-Threading
- Synchronization
- Threads Life Cycle
- Use Cases

17. Essentials of Accessing APIs

- Introduction
- Facebook Messenger
- OpenWeather

Back-end Data Science & Analytics -AI

Intro to Data Science & Analytics

- What is Data Science?
- Difference: Data Science vs Data Analytics vs Machine Learning
- Applications: Healthcare, Finance, Marketing, E-commerce, Social Media
- Data types: Structured vs Unstructured
- Big Data basics and challenges

NumPy for Numerical Data

- Setup: Install & import
- Arrays: Create, index, reshape
- Math: Mean, median, variance, std dev
- Matrix Ops: Dot product, multiplication
- Advanced: Vectorization, numpy.linalg, missing values
- Performance: Optimized computations, memory efficiency

Pandas for Data Analysis

- Why Pandas? Efficient data handling
- Setup: Install & import
- DataFrames: Create, explore, modify
- Clean: Handling missing data, removing duplicates
- Categoricals: Label & One-Hot Encoding
  Analysis: Aggregation, grouping, merging

Matplotlib for Python Plotting

- Intro & install (pip install matplotlib)
- Understand pyplot and basics
- Plots: line, bar, scatter
- Customize: colors, labels, titles
- Add legends, annotations, text
- Customize axes, grid lines
- Use subplot(), subplots()
- Adjust figure size, layout
- 3D plots: use mpl_toolkits.mplot3d

Parallel Pandas for Big Data

- Big Data issues in traditional Pandas
- Intro to Dask & Modin for parallel computing
- Install: pip install dask
- Convert Pandas to Dask DataFrame
- Use lazy execution for scaling
- Clean data: Handle nulls, remove duplicates
- Categorical handling: Label & One-Hot Encoding
- Merge: concat(), merge()
- Group: groupby(), pivot_table()
- Time series: DateTime index, resample,
- rolling calculations

Seaborn for Statistical Plots

- Intro & Advantages: Statistical data visualization
- Setup: Install & import (pip install seaborn)
- Basic Plots: Histogram, KDE, box plot
- Advanced: Violin, swarm, strip plots
- Correlation: Pairplot, heatmaps
- Customize: Themes, color palettes, annotations
- Enhance: Combine with Matplotlib
- Faceting: Multiple plots with FacetGrid
- Regression: Visualize trends with regplot and Implo

Plotly for Dynamic Visualizations

- Intro to Plotly and its uses
- Install: pip install plotly
- Interactive charts: line, bar, scatter
- Hover effects and tooltips
- 3D plots: surface, mesh
- Geospatial: choropleth maps
- Dash: Build web-based dashboards
- Export: save as HTML or image
- Embed plots in web apps

Machine Learning - AI

Back-end Machine Learning - AI

Machine Learning Algorithms

- Intro & Importance
- Types: Supervised, Unsupervised, Reinforcement
- Applications: Healthcare, Finance, E-commerce

Supervised Learning

- Classification: Spam detection, sentiment analysis
- Regression: House prices, sales forecasting
- Key Difference: Classification vs. Regression

Unsupervised Learning

- Clustering: segmentation, anomaly detection
- Dimensionality Reduction: PCA, t-SNE

Reinforcement Learning

- Concepts: Agents, environment, rewards
- Basics: Q-learning

SciKit-Learn for ML Task

- Overview of SciKit-Learn and its features

Data Preprocessing

- Handling missing data
- Encoding categorical variables
- Feature scaling techniques

Model Building

- Splitting data into training and testing sets
- Cross-validation techniques
- Creating machine learning pipelines

Model Evaluation

- Metrics: Accuracy, Precision, Recall, F1-score
- Confusion matrix, ROC-AUC, MSE

Common Algorithms

- Linear and Logistic Regression
- Decision Trees and Random Forest
- K-Nearest Neighbors and Support Vector Machines

SciPy for ML Task

- Overview and difference from NumPy

Scientific Computing

- Linear algebra operations
- Integration and interpolation

Optimization

- Function minimization & cost function optimization

Statistical Functions

- Probability distributions
- Hypothesis testing (T-test, Chi-square test)

K-Means Clustering for ML

- Introduction to clustering and real-world use cases
- Difference between clustering and classification

K-Means Concepts

- Centroid calculation and iterative optimization

Cluster Evaluation Techniques

- Elbow Method
- Silhouette Score

Implementation Aspects

- Using KMeans algorithm
- Handling large datasets with MiniBatchKMeans

Logistic Regression for ML

- Overview and comparison with Linear Regression
- Spam detection and medical diagnosis

Mathematical Foundation

- Sigmoid function and decision boundary

Model Implementation

- Binary and multi-class classification

Model Evaluation

- Precision, Recall, F1-score
- ROC Curve and AUC

Linear Regression for ML

- Applications in prediction problems

Key Assumptions

- Linearity, Independence, Homoscedasticity

Mathematical Concepts

- Ordinary Least Squares (OLS)
  Gradient Descent

Model Implementation

- Single and Multiple Linear Regression
- Model evaluation and prediction

Advanced Topics

- Feature selection and multicollinearity

Random Forest and Decision Trees

Decision Trees

- Concepts and splitting criteria (Gini, Entropy)
  Overfitting and pruning methods

Random Forest

- Concept of ensemble learning
- Bagging and bootstrapping techniques

Implementation Topics

- Building models with Decision Tree and Random Forest algorithms

Back-end Artificial Intelligence (AI)

Introduction to AI

What is AI ?
- Machines mimicking human intelligence
- From sci-fi to reality
- Types of AI: Narrow AI vs. General AI vs. Super AI

AI vs. ML vs. Deep Learning
- Understanding key differences
- Supervised, unsupervised, and reinforcement learning

Applications of AI
- Healthcare: Diagnosis, prediction, surgery
- Finance: Fraud, trading, risk
- E-commerce: Recommendations, chatbots, forecasting
- Automation: Smart homes, cars, robots
- Entertainment: Gaming, music, deepfakes

Ethics in AI
- Fairness, bias, and responsibility
- AI and job displacement
- Privacy and data security concerns
- Regulations and ethical AI practices

Python for AI

Python Recap
- Brush up on basics (variables, loops, functions, OOP)
- Working with data structures (lists, dictionaries, sets)
- File handling and data manipulation
- Writing efficient and readable Python code

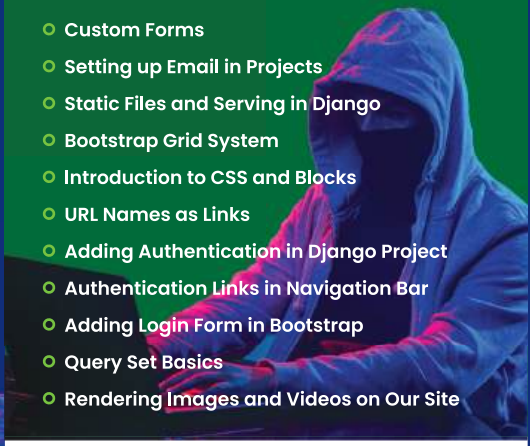
AI Libraries
- NumPy, Pandas: Data handling, data processing
- TensorFlow, Keras: Build AI models, deep learning
- Matplotlib, Seaborn: AI Visualizing outputs
- Scikit-learn: Machine learning algorithms
- NLTK, SpaCy: Natural Language Processing (NLP)

Hands-On Projects
- Create a simple AI chatbot
- Build a spam detection model

Key Skills & Next Steps
- AI Development: Get started
- Model Evaluation: Accuracy, precision, F1-score
- Future AI: Trends ahead

Django

- Introduction to Django
- Difference Between an App and Project
- Understanding Migrations and Their Purpose
- Admin Part: Creating a Superuser in Python
- Views in Django URL Routing
- Render and Relative Import
- Overview of Settings File in Django
- Configuring Templates in Django
- Understanding Models
- Linking Models to Admin
- Creating ModelForms
- Form Validation in Django
- Context in Django
- Forms in a View
- Custom Forms
- Setting up Email in Projects
- Static Files and Serving in Django
- Bootstrap Grid System
- Introduction to CSS and Blocks
- URL Names as Links
- Adding Authentication in Django Project
- Authentication Links in Navigation Bar
- Adding Login Form in Bootstrap
- Query Set Basics
- Rendering Images and Videos on Our Site



Database

- **Introduction to Databases**
  - Overview of SQL (Structured Query Language)
  - Role of databases in various industries
  - MySQL Installation & Setup including Workbench
- **SQL Statements:**
  - Data Definition Language (DDL)
  - Data Manipulation Language (DML)
  - Data Query Language (DQL)
- **Basic Commands:**
  - Create, Insert, Select, Update, Delete, Drop, Alter
- **Constraints:**
  - Primary Key, Foreign Key, Unique, Not Null
- **Operators and Clauses:**
  - IN, Between, Like, Exists and Order BY
- **Joins:**
  - Inner, Left, Right, Full Outer
- **Grouping:**
  - GROUP BY, HAVING, Aggregate Functions
- **Hands-on Schema and Data Creation**

Front-end HTML

1. HTML Basics
- HTML Syntax

2. Text Formatting
- Headings (<h1> to <h6>)
- Paragraphs (<p>)
- Line Breaks (<br>) & Horizontal Rules (<hr>)

3. Lists
- Ordered Lists (<ol>)
- List Items (<li>)
- Un-ordered Lists (<ul>)

4. Links and Anchors
- Hyperlinks (<a>)
- Link Text
- URLs & Paths
- Link Targeting

5. Marquee Element
- Element Introduction & Attributes
- Creating Scrollable Text & Images

6. Images
- Image Tag (<img>)
- Image Source (src Attribute)
- Image Dimensions (Width and Height Attributes)
- Alt Text (alt Attribute)

7. Forms
- Form Tag (<form>)
- Textareas (<textarea>)
- Input Fields (<input>)
- Select Menus (<select>)

8. Div and Spans
- Block-level Elements ( )
- Inline Elements ()

9. Iframe
- Embedding Videos
- Embedding Maps
- Displaying External Content

10. HTML Comments
- Adding Comments

11. CSS Linking
- Linking External CSS (<link>)

12. **CSS Basics** .....
- What is CSS and its role in web development.
- Inline, internal, and external CSS.
- CSS syntax and basic structure.

13. **CSS Selectors** .....
- Applying styles using element, class & ID selec
- Changing text and background colors tors
- Adjusting font properties (size, family, weight)

14. **CSS Borders** .....
- Adjusting border width and color
- Adding rounded corners to elements

15. **CSS Margins and Padding** .....
- Exploring margin properties for spacing control
- Setting margins for individual sides
- Combining Margin and Padding

16. **CSS Fonts** .....
- Customizing text styles with font properties
- Choosing font families & font stacks
- Adjusting font size, weight, style, and color

17. **CSS Tables** .....
- Styling HTML tables and their.elements
- Setting borders and spacing for table cells

18. **CSS** Background & Images .....
- Setting background color and images

19. **JavaScript Basics** .....
- DOM Manipulation
- Functions
- Popup Boxes

React.js

- Module REACT.js
- Overview
- Features of React.js Advantages & Limitations
- Environment Setup
- Install Global Packages
- Add Dependencies & Plugins; Run the Server
- React.js Component
- Rendering Function Component, API State
- Props
- Creating a React Component
- Creating a Function Component
- React.js - Events, Simple Example of Child Events.
- React.js - Redux
- Concept of Redux
- React.js - Router
- Install a React Router; Create Components; Add a Router
- HOOKS in React
- Higher-Order Component
- When to Use Hook; Rules of Hook Usage; Hook State
- Hook Effect

Angular

1. What is Angular?

- Angular version History
- Architecture of Angular Apps
- Setting up the Environmental Setup
- Structure of Angular Projects
- Creating Components Manually
- Creating Components through CMD
- String Interpolation

2. Data Binding, Handling Events & Directives

- String Interpolation vs Property Binding
- Property Binding
- Class Binding
- Style Binding

1. Event Binding
   - (click), (change), (keyup), (keydown)
   - Mouse Events

- Two-Way Binding
- Directives
- Structural Directives (nglf, nglf else, ngFor)
- ngStyle
- ngClass

3. Component Interaction, Input & Output Properties, Pipes, Services & Dependency Injection

1. Component Interaction
   - @Input() and @Output() Decorators
2. Pipes
   - Inbuilt Pipes
   - Creating Custom Pipes
   - @Pipe Decorator

4. HTTP Services and Observables

- What is a Service?
- Dependency Injection in Angular
- Introduction to HTTP
- Error Handling
- Introduction to Observables
- Injectable Decorator
- Connecting to the Backend
- Using HTTP Services

1. CRUD Operations with HTTP
   - GET, POST, PUT, DELETE

5. Routing in Angular

- What is Routing?
- Configuring Routes
- Using the Router-outlet Tag
- Routing Using Router-link Directive
- Programmatic Routing
- Sending Data through Routing
- Lazy Loading Concepts

6. Bootstrap and Template-Driven Forms

- Overview of Bootstrap
- Bootstrap Grid System
- Using the Form Control Class
- Validation of Template-Driven Forms
- ngForm Directive
- ngValue Directive
- ngSubmit Directive
- FormControl

EAT
SLEEP
CODE
REPEAT

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Why Choose This Course?

- **Hands-On Learning**: Real-world projects and datasets
- **Industry-Relevant Skills**: SQL, Python, ML, AI tools
- **Career Boost**: Prep for high-

Get Started Today!

Transform Data into Opportunity

Our Top Courses

- Java Full Stack Development
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- Software Testing
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- C#.NET

Personality Development / Soft Skills

English Communication

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Branch: Chinchwad
Add: KUNAL PLAZA, 2nd Floor, Old Mumbai - Pune Hwy, **Pimpri-Chinchwad**, Maharashtra 411019

Branch: Warje
Add: 1st floor, The Kiran Academy, Dr Ambedkar Chowk, above Royal Car Decor, **Warje** Jakat Naka, Warje, Pune, Maharashtra 411058

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